Answers From Chatgpt

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December 5, 2025

1 Explanation of the SU(2) Global Anomaly Remark

We explain the logic of the remark step by step.

1.1 SU(2) Gauge Theory in Four Dimensions

The weak interaction is described by an $SU(2)$ gauge theory. Therefore, on our four-dimensional spacetime M there is an $SU(2)$ principal bundle

$$P \to M.$$

On a local patch we may treat spacetime as $\mathbb{R}^4$, and a gauge transformation is a map

$$g : \mathbb{R}^4 \to SU(2).$$

1.2 Compactification to S^4 and Large Gauge Transformations

Assume the gauge transformation becomes trivial at spatial infinity:

$$g(x) \to \mathbf{1} \quad \text{as } |x| \to \infty.$$

This allows us to compactify $\mathbb{R}^4$ by adding a point at infinity:

$$\mathbb{R}^4 \cup \{\infty\} \cong S^4,$$

with $g(\infty) = \mathbf{1}$. Hence gauge transformations are classified by maps

$$g : S^4 \to SU(2) \simeq S^3.$$

The homotopy classes of such maps are elements of

$$\pi_4(S^3) \simeq \mathbb{Z}_2 = \{0, 1\}.$$

The nontrivial class (1) corresponds to a "large" gauge transformation.

1.3 SU(2) Doublet Fermions

Fermions charged under the weak force transform in the fundamental (doublet) representation **2** of $SU(2)$. A single such fermion field is a section of the associated vector bundle

$$\mathbf{2} \otimes S \longrightarrow M,$$

where S is the spinor bundle. For our purposes, the only important fact is:

A single SU(2) doublet fermion is one section of a vector bundle.

1.4 Witten's SU(2) Global Anomaly

A key result of Witten (1982) states that under a large gauge transformation $g : S^4 \to S^3$, the partition function of a single $SU(2)$ doublet fermion transforms as

$$Z_{\text{single doublet}} \;\longrightarrow\; (-1)^s \, Z_{\text{single doublet}},$$

where $s = [g] \in \pi_4(S^3) \simeq \mathbb{Z}_2$. Thus

$$s = 0 : \;\; Z \text{ invariant}, \qquad s = 1 : \;\; Z \mapsto -Z.$$

Therefore:

A single SU(2) doublet suffers from a global SU(2) anomaly.

1.5 Multiple Doublets and Anomaly Cancellation

If there are N doublets, the total transformation is

$$Z \longrightarrow (-1)^{Ns} Z.$$

The theory is gauge invariant for all gauge transformations if and only if N is even. In the Standard Model, one generation contains

$$3 \text{ quark doublets } + 1 \text{ lepton doublet } = 4 \text{ doublets.}$$

Hence

$$(-1)^{4s} = +1,$$

and the anomaly cancels.

This explains, for instance, why one cannot simply change $SU(3)$ color to $SU(4)$: it would change the number of weak doublets, potentially making N odd and producing an inconsistency.

1.6 Summary

- Large $SU(2)$ gauge transformations are classified by $\pi_4(S^3) \simeq \mathbb{Z}_2$.
- A single doublet fermion changes sign under the nontrivial transformation.
- A consistent theory must have an *even* number of doublets.
- The Standard Model satisfies this automatically ($N = 4$ per generation).

2 Fiber Choice in the Covering Map $\mathbb{R} \to S^1$

Consider the covering map

$$p : \mathbb{R} \longrightarrow S^1, \qquad p(x) = e^{2\pi i x}.$$

For any point $z = e^{2\pi i\alpha} \in S^1$, the fiber is

$$p^{-1}(z) = \{x \in \mathbb{R} \mid e^{2\pi i x} = z\} = \alpha + \mathbb{Z},$$

a translated copy of $\mathbb{Z}$. Thus every fiber is homeomorphic to the discrete set $\mathbb{Z}$.

In the long exact sequence of a fibration, one chooses a basepoint $b_0 \in S^1$ and defines the fiber as $p^{-1}(b_0)$. Although $b_0 = 1 \in S^1$ (the identity element) is the conventional choice, any b_0 works because S^1 is path-connected and homotopy groups based at different points are naturally isomorphic.

3 Homotopy Groups of the Discrete Space $\mathbb{Z}$

Although $\mathbb{Z}$ is not a manifold, it is a valid topological space and its homotopy groups are well-defined.

3.1 All Continuous Maps $S^n \to \mathbb{Z}$ Are Constant ($n \geq 1$)

Since $\mathbb{Z}$ is discrete, each singleton $\{k\} \subset \mathbb{Z}$ is open. A continuous map from a connected space S^n to a discrete space must have connected image, hence the image consists of only one point. Thus any map

$$f : S^n \to \mathbb{Z}, \quad n \geq 1$$

is constant.

3.2 Computation of $\pi_n(\mathbb{Z})$

If we fix the basepoint $k_0 \in \mathbb{Z}$, the only based map $S^n \to \mathbb{Z}$ is the constant map $f(x) \equiv k_0$. Hence, for $n \geq 1$,

$$\pi_n(\mathbb{Z}, k_0) = 0.$$

For $n = 0$, the path components of $\mathbb{Z}$ are just its points, so

$$\pi_0(\mathbb{Z}) \cong \mathbb{Z}.$$

3.3 Summary

$$\boxed{\pi_0(\mathbb{Z}) = \mathbb{Z}, \qquad \pi_{n\geq 1}(\mathbb{Z}) = 0.}$$

3.4 Application: Computing $\pi_n(S^1)$ Using the Covering Space

From the fibration

$$\mathbb{Z} \longrightarrow \mathbb{R} \longrightarrow S^1,$$

the long exact sequence gives

$$\pi_n(S^1) \cong \pi_{n-1}(\mathbb{Z}).$$

Using our calculation of $\pi_{n-1}(\mathbb{Z})$, we obtain the classical result:

$$\pi_n(S^1) = \begin{cases} \mathbb{Z}, & n = 1, \\ 0, & n \neq 1. \end{cases}$$

4 Understanding why $\partial : \mathbb{Z} \to \mathbb{Z}$ must be multiplication by 2

Consider the fibration

$$SO(2) \;\longrightarrow\; SO(3) \;\longrightarrow\; S^2.$$

The long exact sequence in homotopy (for $n = 3$ in the book's notation) contains the segment

$$\pi_2(SO(2)) \longrightarrow \pi_2(SO(3)) \longrightarrow \pi_2(S^2) \xrightarrow{\partial} \pi_1(SO(2)) \longrightarrow \pi_1(SO(3)) \longrightarrow \pi_1(S^2).$$

We substitute the known homotopy groups:

$$\pi_2(SO(2)) = 0, \quad \pi_2(S^2) = \mathbb{Z}, \quad \pi_1(SO(2)) = \mathbb{Z}, \quad \pi_1(SO(3)) = \mathbb{Z}_2, \quad \pi_1(S^2) = 0.$$

Thus the exact sequence becomes

$$0 \longrightarrow \pi_2(SO(3)) \longrightarrow \mathbb{Z} \xrightarrow{\partial} \mathbb{Z} \xrightarrow{\text{mod } 2} \mathbb{Z}_2 \longrightarrow 0.$$

4.1 Step 1: Determining the form of ∂

Exactness at $\pi_1(SO(2)) = \mathbb{Z}$ means

$$\mathrm{Im}(\partial) = \ker(\mathrm{mod} 2).$$

Since

$$\ker(\text{mod}2) = 2\mathbb{Z},$$

we obtain

$$\text{Im}(\partial) = 2\mathbb{Z}.$$

Any group homomorphism $\partial : \mathbb{Z} \to \mathbb{Z}$ must have the form

$$\partial(n) = kn, \qquad k \in \mathbb{Z}.$$

Its image is $k\mathbb{Z}$. To satisfy $\text{Im}(\partial) = 2\mathbb{Z}$, the only possibility is

$$k = \pm 2.$$

Hence

$$\partial(n) = \pm 2n.$$

This is the meaning of the sentence in the book:

> "This forces $\partial : \mathbb{Z} \to \mathbb{Z}$ to be a multiplication by 2."

4.2 Step 2: Computing $\pi_2(SO(3))$

Exactness at $\pi_2(S^2) = \mathbb{Z}$ implies

$$\text{Im}\Big(\pi_2(SO(3)) \longrightarrow \mathbb{Z}\Big) = \ker(\partial).$$

Since $\partial(n) = \pm 2n$, its kernel is

$$\ker(\partial) = \{0\}.$$

Thus

$$\text{Im}\Big(\pi_2(SO(3)) \to \mathbb{Z}\Big) = 0.$$

But the map $0 \to \pi_2(SO(3))$ at the beginning of the sequence is injective, so the only way $\pi_2(SO(3))$ can map to 0 inside $\mathbb{Z}$ is if

$$\pi_2(SO(3)) = 0.$$

So the conclusion is:

$$\boxed{\pi_2(SO(3)) = 0.}$$

5 Unitarity in 2-2 scattering

The partial–wave coefficients are defined by projecting the T–matrix onto two–particle states of definite total momentum and total angular momentum,

$$f_J(p^2) \;\propto\; \frac{\langle\, p, J, \vec{m} \,|\, \hat{T} \,|\, p, J, \vec{m} \,\rangle}{\langle\, p, J, \vec{m} \,|\, p, J, \vec{m} \,\rangle}\,. \tag{1}$$

Because $\hat{T}$ is invariant under translations and under $SO(1, d-1)$, the matrix element in this basis is diagonal in $(J, \vec{m})$, so the proportionality constant is independent of p^2 and of $\vec{m}$. The result therefore depends only on the Lorentz–invariant combination $p^2 = s$.

We start from the angular form of the elastic unitarity equation,

$$2\,\Im T(s,t(z)) = \int_{-1}^{1} dz'\, w(z') \int_{-1}^{1} dz''\, w(z'')\, \mathcal{P}_d(z,z',z'')\, T(s,t(z'))\, T^*(s,t(z'')), \tag{2}$$

where

$$w(z) = (1-z^2)^{\frac{d-4}{2}}.$$

Next we insert the partial-wave expansion,

$$T(s,t(z)) = \sum_{L=0}^{\infty} n_L^{(d)}\, f_L(s)\, P_L^{(d)}(z), \tag{3}$$

and substitute this expression into (2). At this stage we obtain

$$\begin{aligned} 2\,\Im T(s,t(z)) = \sum_{L,L'} n_L^{(d)} n_{L'}^{(d)} f_L(s) f_{L'}^*(s) \int_{-1}^{1} dz'\, w(z')\, \mathcal{P}_d(z,z',z'')\, P_L^{(d)}(z') \\ \times \int_{-1}^{1} dz''\, w(z'')\, P_{L'}^{(d)}(z''). \end{aligned} \tag{4}$$

To isolate the coefficient $f_J(s)$ we project both sides of (4) onto the J-th partial wave, multiplying by

$$\frac{\mathcal{N}_d}{2}\, w(z)\, P_J^{(d)}(z)$$

and integrating over z. Using the projection formula

$$f_J(s) = \frac{\mathcal{N}_d}{2} \int_{-1}^{1} dz\, w(z)\, P_J^{(d)}(z)\, T(s,t(z)), \tag{5}$$

the left-hand side becomes simply

$$\frac{\mathcal{N}_d}{2} \int_{-1}^{1} dz\, w(z) P_J^{(d)}(z)\, 2\,\Im T(s,t(z)) = 2\,\Im f_J(s).$$

Now we insert the kernel decomposition (eq. (67))

$$\mathcal{P}_d(z,z',z'') = \frac{1}{2} \mathcal{N}_d^2\, w(z')\, w(z'') \sum_{L=0}^{\infty} n_L^{(d)}\, P_L^{(d)}(z)\, P_L^{(d)}(z')\, P_L^{(d)}(z''), \tag{6}$$

and use the orthogonality relation

$$\frac{1}{2} \int_{-1}^{1} dz\, w(z)\, P_J^{(d)}(z)\, P_L^{(d)}(z) = \frac{\delta_{JL}}{\mathcal{N}_d n_J^{(d)}}. \tag{7}$$

Substituting (6) into (4) and performing the two z', z'' integrals, only the term $L = L' = J$ survives due to (7), giving

$$\frac{\mathcal{N}_d}{2} \int dz\, w(z) P_J^{(d)}(z)\, (\text{RHS}) = \frac{(s-4m^2)^{\frac{d-3}{2}}}{\sqrt{s}}\, |f_J(s)|^2.$$

Equating both sides yields the elastic partial-wave unitarity condition

$$2\,\Im f_J(s) = \frac{(s-4m^2)^{\frac{d-3}{2}}}{\sqrt{s}}\, |f_J(s)|^2, \tag{8}$$

which is eq. (68) of the main text.

6 Solution Structure of S_i from Given Helicity Data

We are given three helicities

$$h_1,\, h_2,\, h_3 \in \tfrac{1}{2}\mathbb{Z}, \qquad h_1 + h_2 + h_3 > 0, \tag{1}$$

three spins

$$S_i \geq 0$$

(half-integers), and integers

$$k_i \geq 0$$

satisfying

$$S_i - k_i = h_i, \qquad i = 1, 2, 3. \tag{2}$$

We also impose the constraint

$$k_1 + k_2 + k_3 \;\leq\; 2S_{\min}, \qquad S_{\min} := \min(S_1, S_2, S_3). \tag{3}$$

Elementary implications

From (2) and $k_i \geq 0$,

$$S_i = h_i + k_i \quad \Rightarrow \quad S_i \geq h_i. \tag{4}$$

Hence every spin satisfies the effective lower bound

$$S_i \geq \max(0, h_i). \tag{5}$$

Let the ordered helicities be

$$h_{(1)} \leq h_{(2)} \leq h_{(3)}. \tag{6}$$

Then

$$S_{\min} = \min_i S_i \;\geq\; \min_i \max(0, h_i) = \max(0, h_{(1)}). \tag{7}$$

Rewriting the constraint

Using $k_i = S_i - h_i$ in (3),

$$k_1 + k_2 + k_3 = (S_1 + S_2 + S_3) - (h_1 + h_2 + h_3) \leq 2S_{\min}. \tag{8}$$

From $S_i \geq S_{\min}$, we have

$$k_i = S_i - h_i \geq S_{\min} - h_i \tag{9}$$

and at the same time $k_i \geq 0$, so we obtain

$$k_i \geq \max\{0, S_{\min} - h_i\} \tag{10}$$

Define

$$f(S_{\min}) := \sum_{i=1}^{3} \max\{0,\, S_{\min} - h_i\}. \tag{11}$$

This quantity is the minimal possible value of $k_1 + k_2 + k_3$ at fixed $S_{\min}$.
Thus a necessary and sufficient condition for feasible k_i is

$$f(S_{\min}) \leq 2S_{\min}. \tag{12}$$

With ordered helicities,

$$f(S_{\min}) = \begin{cases} 0, & 0 \leq S_{\min} \leq h_{(1)}, \\ S_{\min} - h_{(1)}, & h_{(1)} \leq S_{\min} \leq h_{(2)}, \\ 2S_{\min} - (h_{(1)} + h_{(2)}), & h_{(2)} \leq S_{\min} \leq h_{(3)}, \\ 3S_{\min} - (h_1 + h_2 + h_3), & S_{\min} \geq h_{(3)}. \end{cases} \tag{13}$$

Imposing (12) yields:

- Region I:
  $$0 \leq S_{\min} \leq h_{(1)}$$
  Always satisfied.

- Region II:
  $$h_{(1)} \leq S_{\min} \leq h_{(2)}$$
  $$S_{\min} - h_{(1)} \geq 0, \quad S_{\min} - h_{(2)} \leq 0, \quad S_{\min} - h_{(3)} \leq 0 \tag{14}$$
  So we have
  $$f(S_{\min}) = S_{\min} - h_{(1)} \leq 2S_{\min} \quad \Rightarrow \quad S_{\min} \geq -h_{(1)}. \tag{15}$$
  If $h_{(1)} < 0$, it will give us a non-trival lower bound.

- Region III:
  $$h_{(2)} \leq S_{\min} \leq h_{(3)}$$
  $$S_{\min} - h_{(1)} \geq 0, \quad S_{\min} - h_{(2)} \geq 0, \quad S_{\min} - h_{(3)} \leq 0 \tag{16}$$
  So we have
  $$f(S_{\min}) = 2S_{\min} - (h_{(1)} + h_{(2)}) \leq 2S_{\min} \quad \Rightarrow \quad h_{(1)} + h_{(2)} \geq 0. \tag{17}$$

- Region IV:
  $$S_{\min} \geq h_{(3)}$$
  $$3S_{\min} - (h_1 + h_2 + h_3) \leq 2S_{\min} \quad \Rightarrow \quad S_{\min} \leq h_1 + h_2 + h_3. \tag{18}$$

Combining (7) with the former constraints, a solution exists iff

$$h_{(1)} + h_{(2)} \geq 0, \tag{19}$$

and the interval

$$S_{\min} \in \Big[\max\{0, h_{(1)}\},\ h_1 + h_2 + h_3\Big] \tag{20}$$

is nonempty.
Since

$$h_1 + h_2 + h_3 > 0,$$

the upper bound is always positive.
Thus (19) is the essential obstruction that rules out forbidden helicity configurations.

Whenever a valid $S_{\min}$ exists, the spins are reconstructed by

$$S_i = h_i + k_i, \qquad k_i \in \mathbb{Z}_{\geq 0}, \tag{21}$$

with

$$k_i \geq \max(0,\ S_{\min} - h_i), \qquad k_1 + k_2 + k_3 \leq 2S_{\min}. \tag{22}$$

Because both

$$S_{\min}$$

and

$$k_1 + k_2 + k_3$$

are bounded, only finitely many triples

$$(S_1, S_2, S_3)$$

are possible for specific helicities.

6.1 A general lower bound valid for all solutions for $h_1 < 0, h_{2,3} \geq 0$

We do not assume which S_i equals $S_{\min}$. We only use the relations

$$S_i = h_i + k_i, \qquad k_i \geq 0, \tag{23}$$

and the definition of the minimal spin

$$S_i \geq S_{\min}. \tag{24}$$

From this inequality we immediately obtain

$$S_{\min} - h_i \ \leq\ S_i - h_i = k_i, \tag{25}$$

and therefore

$$k_i \ \geq\ S_{\min} - h_i. \tag{26}$$

Combining with $k_i \geq 0$ gives the unified inequality

$$k_i \;\geq\; \max\{0,\, S_{\min} - h_i\}, \qquad i = 1, 2, 3. \tag{27}$$

Summing these three inequalities yields

$$k_1 + k_2 + k_3 \;\geq\; \sum_{i=1}^{3} \max\{0,\, S_{\min} - h_i\}. \tag{28}$$

Now consider the case of exactly one negative helicity and two positive helicities. After reordering,

$$h_1 < 0 < h_2 \leq h_3. \tag{29}$$

Since each $S_i \geq 0$, we necessarily have

$$S_{\min} \geq 0. \tag{30}$$

We determine the signs of the terms in the previous inequality:

- For $i = 1$: since $h_1 < 0$ and $S_{\min} \geq 0$, we have
$$\max\Big(0, S_{\min} - h_1\Big) = S_{\min} - h_1 > 0. \tag{31}$$

- For $i = 2, 3$: since $h_2, h_3 > 0$, we only need the trivial bounds
$$\max\Big(0, S_{\min} - h_2\Big) \geq 0, \qquad \max\Big(0, S_{\min} - h_3\Big) \geq 0. \tag{32}$$

Therefore,

$$k_1 + k_2 + k_3 \;\geq\; (S_{\min} - h_1) + 0 + 0 = S_{\min} - h_1. \tag{33}$$

Combining this with the fundamental inequality

$$k_1 + k_2 + k_3 \leq 2S_{\min}, \tag{34}$$

we obtain the chain

$$S_{\min} - h_1 \;\leq\; k_1 + k_2 + k_3 \;\leq\; 2S_{\min} \quad\Longrightarrow\quad S_{\min} - h_1 \leq 2S_{\min}. \tag{35}$$

Rearranging gives the universal lower bound

$$-h_1 \leq S_{\min} \quad\Longrightarrow\quad S_{\min} \geq |h_1|. \tag{36}$$

Since $S_1 \geq S_{\min}$ for all solutions, we also have

$$S_1 \geq S_{\min} \geq |h_1|. \tag{37}$$

Thus we have established the following conclusion:

$$\boxed{\text{For any admissible solution, we necessarily have } S_1 \geq |h_1|.} \tag{38}$$

This bound required no assumption about which spin equals $S_{\min}$; it is a universal constraint following only from the defining inequalities.

Existence of a solution saturating the lower bound
Assume in addition the necessary helicity condition

$$h_1 + h_2 \geq 0 \qquad \Longleftrightarrow \qquad h_2 \geq |h_1|, \tag{39}$$

which is required for the existence of any solution.
We now construct an explicit solution saturating the inequality $S_1 \geq |h_1|$. Define

$$S_1 = |h_1|, \qquad S_2 = h_2, \qquad S_3 = h_3. \tag{40}$$

Using $S_i = h_i + k_i$ we obtain the corresponding integers

$$k_1 = S_1 - h_1 = |h_1| - h_1 = 2|h_1|, \qquad k_2 = 0, \qquad k_3 = 0. \tag{41}$$

Since $h_2 \geq |h_1|$ and $h_3 \geq h_2$, we have

$$S_2 = h_2 \geq |h_1|, \qquad S_3 = h_3 \geq h_2 \geq |h_1|, \tag{42}$$

so that in this configuration

$$S_{\min} = S_1 = |h_1|. \tag{43}$$

Finally,

$$k_1 + k_2 + k_3 = 2|h_1| = 2S_{\min}, \tag{44}$$

which saturates the inequality $k_1 + k_2 + k_3 \leq 2S_{\min}$.
Thus the lower bound is sharp: the minimal possible value of S_1 is

$$\boxed{S_1^{\min} = |h_1|.} \tag{45}$$

An explicit solution attaining this minimum is

$$(S_1, S_2, S_3) = \Big(|h_1|,\ h_2,\ h_3\Big), \qquad (k_1, k_2, k_3) = \Big(2|h_1|,\ 0,\ 0\Big). \tag{46}$$

This completes the demonstration that, in the case of one negative helicity and two positive helicities with $h_2 \geq |h_1|$, we not only have the universal bound $S_1 \geq |h_1|$ but also a concrete solution saturating it.